MSc. Project Report: Survey on Equations over Finite Fields

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Abstract

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Preface

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Acknowledgements

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List of Notations

The following table lists the notation used throughout this report, organized by topic.

Symbol	Meaning
General	
$\mathbb{N}$	Set of all natural numbers i.e., positive integers
$\mathbb{Z}$	Set of all integers
$\mathbb{Z}_0$	Set of all non-negative integers
$\mathbb{R}$	Set of real numbers
$\mathbb{C}$	Set of complex numbers
$[n]$	The set $\{1, 2, \ldots, n\}$ for $n \in \mathbb{N}$
i	Imaginary unit: $i^2 = -1$
S^1	$\{z \in \mathbb{C} : \lvert z\rvert = 1\}$, the multiplicative group of unit complex numbers
ζ_m	A primitive m-th root of unity
$[x]$	Floor function: largest integer $\leq x$; also written $\lfloor x \rfloor$
Finite Fields	
$\mathbb{F}_q$	Finite field with q elements
$\mathbb{F}_q^*$	Multiplicative group of nonzero elements of $\mathbb{F}_q$
$\mathbb{F}_p$	Prime field with p elements, $p = \mathrm{char}(\mathbb{F}_q)$
$\mathbb{F}_q^{*\,(d)}$	Subgroup of d-th powers in $\mathbb{F}_q^*$
E, F, L, K	Generic notation for finite fields or finite extensions of $\mathbb{F}_q$
$[E : \mathbb{F}_q]$	Degree of the extension $E/\mathbb{F}_q$
$\mathbb{F}_q[X_1, \ldots, X_n]$	Polynomial ring in n variables over $\mathbb{F}_q$
$\mathbb{F}_q^{\leq d}[X_1, \ldots, X_n]$	Polynomials in $\mathbb{F}_q[X_1, \ldots, X_n]$ of total degree at most d
Characters of Finite Abelian Groups	
G	A finite Abelian group with identity element 1_G
Z_k	Cyclic multiplicative group of order k
χ_0	Trivial character of G: $\chi_0(g) = 1$ for all $g \in G$
$\overline{\chi}$	Conjugate (and inverse) character of χ: $\overline{\chi}(g) = \overline{\chi(g)}$
$\mathrm{Hom}(A, B)$	Set of all group homomorphisms from A to B
$\widehat{G} = \mathrm{Hom}(G, S^1)$	Dual group of G; the group of all characters of G
$\mathrm{Ann}(\chi)$	Annihilator of character χ: $\{g \in G : \chi(g) = 1\}$, a subgroup of G
$\mathrm{Ann}(H)$	Annihilator of subgroup $H \leq G$: $\{\chi \in \widehat{G} : \chi(h) = 1\ \forall h \in H\}$

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Symbol	Meaning
ev_g	Evaluation character of $\widehat{G}$ defined by $\mathrm{ev}_g(\chi) = \chi(g)$
Maps on Finite Fields	
$\mathrm{Tr}\colon \mathbb{F}_q \to \mathbb{F}_p$	Absolute trace function
$N\colon \mathbb{F}_q \to \mathbb{F}_p$	Absolute norm function
$\mathrm{Tr}_{E/\mathbb{F}_q}$	Trace function from E to $\mathbb{F}_q$
$\mathrm{N}_{E/\mathbb{F}_q}$	Norm function from E to $\mathbb{F}_q$
$\sigma\colon \mathbb{F}_q \to \mathbb{F}_q$	Frobenius automorphism: $\sigma(\alpha) = \alpha^p$
Additive Characters of $\mathbb{F}_q$	
χ, τ	Generic notation for additive characters
χ_1	Canonical additive character of $\mathbb{F}_q$: $\chi_1(\alpha) = e^{2\pi i\,\mathrm{Tr}(\alpha)/p}$
χ_β	Additive character defined by $\chi_\beta(\alpha) = \chi_1(\beta\alpha)$, for $\beta \in \mathbb{F}_q$
$\boldsymbol{\mathcal{X}}_q$	Set of all additive characters of $\mathbb{F}_q$
$\chi^{(r)}$	Additive character of $\mathbb{F}_q$ lifted to E via $\chi^{(r)} = \chi \circ \mathrm{Tr}_{E/\mathbb{F}_q}$, $[E : \mathbb{F}_q] = r$
Multiplicative Characters of $\mathbb{F}_q$	
ψ, λ	Generic notation for multiplicative characters
ψ_0	Trivial multiplicative character of $\mathbb{F}_q$
η	Quadratic character of $\mathbb{F}_q$ (for odd q): $\eta(\alpha) = \pm 1$ according as α is a square
$\boldsymbol{\mathcal{M}}_q$	Set of all multiplicative characters of $\mathbb{F}_q^*$
$\boldsymbol{\mathcal{M}}_q^{(e)}$	Set of multiplicative characters of $\mathbb{F}_q$ of exponent e
$\mathrm{ord}(\psi)$	Order of ψ: smallest positive integer d with $\psi^d = \psi_0$
$\left(\frac{\alpha}{q}\right)$	Legendre symbol; equals $\eta(\alpha)$ for $\alpha \in \mathbb{F}_q$
$\psi^{(r)}$	Multiplicative character of $\mathbb{F}_q$ lifted to E via $\psi^{(r)} = \psi \circ \mathrm{N}_{E/\mathbb{F}_q}$, $[E\colon \mathbb{F}_q] = r$
Gaussian and Jacobi Sums	
$G(\psi, \chi)$	Gaussian sum: $\sum_{\alpha \in \mathbb{F}_q^*} \psi(\alpha)\,\chi(\alpha)$, for $\psi \in \boldsymbol{\mathcal{M}}_q$, $\chi \in \boldsymbol{\mathcal{X}}_q$
$J_\alpha(\lambda_1, \ldots, \lambda_k)$	Generalized Jacobi sum: $\displaystyle\sum_{\substack{\alpha_1,\ldots,\alpha_k \in \mathbb{F}_q \\ \alpha_1+\cdots+\alpha_k=\alpha}} \lambda_1(\alpha_1)\cdots\lambda_k(\alpha_k)$
$J(\lambda_1, \ldots, \lambda_k)$	Jacobi sum; special case $\alpha = 1$ of J_α
$J_0(\lambda_1, \ldots, \lambda_k)$	Jacobi sum at zero; special case $\alpha = 0$ of J_α
Kloosterman Sums	
$K(\chi; a, b)$	Kloosterman sum: $\sum_{\gamma \in \mathbb{F}_q^*} \chi(a\gamma + b\gamma^{-1})$, for $\chi \in \boldsymbol{\mathcal{X}}_q$, $a, b \in \mathbb{F}_q$
$K(\chi^{(r)}; a, b)$	Kloosterman sum lifted to the degree-r extension of $\mathbb{F}_q$
w_1, w_2	Reciprocal roots of the L-function associated to a Kloosterman sum
Salié and Jacobsthal Sums	
$S(\chi, \tau)$	Salié sum: $\sum_{\gamma \in \mathbb{F}_q^*} \eta(\gamma)\,\chi(\gamma)\,\tau(\gamma^{-1})$, for $\chi, \tau \in \boldsymbol{\mathcal{X}}_q$, q odd
$H_n(\alpha)$	Jacobsthal sum: $\sum_{\gamma \in \mathbb{F}_q} \eta(\gamma^{n+1} + \alpha\gamma)$, for $\alpha \in \mathbb{F}_q^*$

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Symbol	Meaning
$I_n(\alpha)$	Associated Jacobsthal sum: $\sum_{\gamma\in\mathbb{F}_q} \eta(\gamma^n+\alpha)$, for $\alpha \in \mathbb{F}_q^*$
Classical Riemann Zeta Function	
$\zeta(s)$	Riemann zeta function: $\sum_{n=1}^{\infty} n^{-s}$, for $\Re(s) > 1$
$\xi(s)$	Completed zeta function: $\frac{1}{2}s(s-1)\,\Gamma(s/2)\,\pi^{-s/2}\,\zeta(s)$
$\Gamma(s)$	Gamma function: $\int_0^\infty e^{-t}t^{s-1}\,dt$
$\Lambda(n)$	Von Mangoldt function: $\log p$ if $n = p^k$, else 0
ρ	Non-trivial zero of $\xi(s)$
Dirichlet L-functions over $\mathbb{F}_q[X]$	
Φ	Set of all monic polynomials in $\mathbb{F}_q[X]$
Φ_d	Monic polynomials in $\mathbb{F}_q[X]$ of degree exactly d
$\mathrm{Irr}(\Phi)$	Set of irreducible monic polynomials in Φ
$\mathfrak{n}(f) = q^{\deg(f)}$	Norm of the polynomial $f \in \mathbb{F}_q[X]$
G	Group of rational functions $r = h_1/h_2$ in $\mathbb{F}_q(X)$ with $h_1, h_2 \in \Phi$ monic
$\overline{G}$	Subgroup of G on which a given character is defined; extended to G by zero
$L(s,\chi)$	Dirichlet L-function: $\sum_{h\in\Phi} \chi(h)\,\mathfrak{n}(h)^{-s}$, for a multiplicative character χ
$\overline{L}(z,\chi)$	Power series satisfying $\overline{L}(z,\chi) = \sum_{h\in\Phi} \chi(h)\,z^{\deg(h)}$ and $\overline{L}(q^{-s},\chi) = L(s,\chi)$
L_k	k-th log-derivative coefficient: $\displaystyle\sum_{\substack{h\in\mathrm{Irr}(\Phi)\\ \deg(h)\mid k}} \deg(h)\,\chi(h)^{k/\deg(h)}$
Equations over $\mathbb{F}_q$ *— Solution Sets*	
$\mathcal{V}(f)$	Solution set of $f(X_1,\ldots,X_n) = 0$ in $\mathbb{F}_q^n$
$Z(f)$	Number of solutions: $\lvert\mathcal{V}(f)\rvert$
$\overline{\mathcal{V}}(f)$	Set of *non-trivial* solutions (excluding $(0,\ldots,0)$ if it's a zero) of $f = 0$
$\overline{Z}(f)$	Number of non-trivial solutions: $\lvert\overline{\mathcal{V}}(f)\rvert$
$\mathcal{V}(f_1,\ldots,f_k)$	Common solution set: $\mathcal{V}(f_1)\cap\cdots\cap\mathcal{V}(f_k)$
$Z(f_1,\ldots,f_k)$	Count of common solutions: $\lvert\mathcal{V}(f_1,\ldots,f_k)\rvert$
$Z_E(P=0)$	Number of solutions of $P(X_1,\ldots,X_n) = 0$ in E where $P \in \mathbb{F}_q[X_1,\ldots,X_n]$, $[E\colon \mathbb{F}_q] > 1$
Ω_d	Space $\mathbb{F}_q^{\leq d}[X_1,\ldots,X_n]$ of polynomials of total degree at most d
$\omega(d)$	Set of n-tuples $(i_1,\ldots,i_n) \in \mathbb{Z}_0^n$ with $i_1+\cdots+i_n \leq d$
Quadratic Forms	
C_f	Coefficient matrix of a quadratic form f: $(C_f)_{ij} = a_{ij}$ where $f = X^\top C_f X$
$\det(f)$	Determinant of the coefficient matrix C_f
$\vartheta : \mathbb{F}_q \to \mathbb{Z}$	Auxiliary function: $\vartheta(0) = q-1$ and $\vartheta(\alpha) = -1$ for $\alpha \neq 0$
Diagonal Equations	
$d_i = \gcd(k_i, q-1)$	Exponent reduced modulo the order of $\mathbb{F}_q^*$, for $a_iX_i^{k_i}$ in a diagonal equation
$M(d_1,\ldots,d_n)$	$\#\left\{(t_1,\ldots,t_n) \in \prod_i [d_i-1] : \frac{t_1}{d_1}+\cdots+\frac{t_n}{d_n} \in \mathbb{Z}\right\}$
$D = \mathrm{lcm}(d_1,\ldots,d_n)$	LCM of exponents; appears in the closed-form formula for $M(d_1,\ldots,d_n)$
Character Sums with Polynomials (Weil Sums)	

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Symbol	**Meaning**
$W(\chi; f)$	Additive Weil sum: $\sum_{\alpha \in \mathbb{F}_q} \chi(f(\alpha))$, for $\chi \in \boldsymbol{\mathcal{X}}_q$, $f \in \mathbb{F}_q[X]$
$W(\psi; f)$	Multiplicative Weil sum: $\sum_{\alpha \in \mathbb{F}_q} \psi(f(\alpha))$, for $\psi \in \boldsymbol{\mathcal{M}}_q$, $f \in \mathbb{F}_q[X]$
$W(\psi, \chi; f, g)$	Mixed Weil sum: $\sum_{\alpha \in \mathbb{F}_q} \psi(f(\alpha))\, \chi(g(\alpha))$, for $\psi \in \boldsymbol{\mathcal{M}}_q$, $\chi \in \boldsymbol{\mathcal{X}}_q$, $f, g \in \mathbb{F}_q[X]$
Lifted Weil Sums (over Extension Fields)	
$E = \mathbb{F}_{q^k}$	Degree-k extension of $\mathbb{F}_q$; $[E : \mathbb{F}_q] = k$
$W^{(k)}(\psi^{(k)}; f)$	Lifted multiplicative Weil sum: $\sum_{\alpha \in E} \psi^{(k)}(f(\alpha))$
$W^{(k)}(\chi^{(k)}; g)$	Lifted additive Weil sum: $\sum_{\alpha \in E} \chi^{(k)}(g(\alpha))$
$W^{(k)}(\psi^{(k)}, \chi^{(k)}; f, g)$	Lifted mixed Weil sum: $\sum_{\alpha \in E} \psi^{(k)}(f(\alpha))\, \chi^{(k)}(g(\alpha))$
Special L-functions (Weil chapter)	
$\overline{G}$ (Weil context)	The specific subgroup $\{r \in G : h_1(\gamma_i) \cdot h_2(\gamma_i) \neq 0\ \forall i\}$ for roots γ_i of a fixed f
$\varsigma : \overline{G} \to \mathbb{F}_q$	Multiplicative map $\varsigma(r) = \prod_i f(\alpha_i) \cdot \prod_j f(\beta_j)^{-1}$ for $r = \prod(X - \alpha_i)/\prod(X - \beta_j)$
$\varrho : G \to \mathbb{F}_q$	Additive map $\varrho(r) = \sum_i g(\alpha_i) - \sum_j g(\beta_j)$ for a fixed polynomial g
$\xi : G \to \mathbb{C}$	Combined character: $\xi(r) = (\psi \circ \varsigma)(r) \cdot (\chi \circ \varrho)(r)$
$L(s, \xi)$	L-function on ξ: $\sum_{h \in \Phi} \xi(h)\, \mathfrak{n}(h)^{-s}$
$\overline{L}(z, \xi)$	Power series: $\sum_{h \in \Phi} \xi(h)\, z^{\deg(h)}$, related by $\overline{L}(q^{-s}, \xi) = L(s, \xi)$
$L^{(k)}(s, \xi^{(k)})$	L-function on the lifted character $\xi^{(k)}$ over $\mathbb{F}_{q^k}$
$\overline{L}^{(k)}(z, \xi^{(k)})$	Power series form of the lifted L-function
$\Phi^{(k)}$	Set of monic polynomials over $E = \mathbb{F}_{q^k}$
$\mathfrak{n}^{(k)}(h) = q^{k \cdot \deg(h)}$	Norm of $h \in \Phi^{(k)}$ over E; analog of $\mathfrak{n}(h) = q^{\deg(h)}$ for $\mathbb{F}_q$
Absolute Varieties	
$\mathcal{A}$	Variety defined over $\mathbb{F}_q$
$Z_k(\mathcal{A})$	Number of $\mathbb{F}_{q^k}$-rational points of $\mathcal{A}$: $\lvert\{(\alpha_1, \ldots, \alpha_n) \in \mathcal{A} : \alpha_i \in \mathbb{F}_{q^k}\}\rvert$

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Chapter 1

Decoding from Character Evaluations

Character sums of polynomials over finite fields carry deep arithmetic information, and one of the most striking manifestations of this is that they give rise to good error-correcting codes. The mechanism is simple: two polynomials whose character evaluations are close must be nearly indistinguishable by the character, and the Weil bounds make this precise. For a multiplicative character ψ of $\mathbb{F}_q$ and two polynomials $f, g \in \mathbb{F}_q[X]$ of small degree, the Weil bound controls gives a lower bound on the Hamming distance between the codewords $(\psi(f(\alpha)))_{\alpha \in \mathbb{F}_q}$ and $(\psi(g(\alpha)))_{\alpha \in \mathbb{F}_q}$. The resulting codes turn out to have distance approaching $q/2$ while the message length grows polynomially in q.

Two families are of particular interest. For q odd, applying the quadratic residue character $\psi(\beta) = \beta^{(q-1)/2}$ to squarefree monic polynomials $g(X)$ of degree $d = o(\sqrt{q})$ defines a nonlinear code over $\{0, \pm 1\}$ of length q. For $q = 2^b$, taking $g(X)$ with only odd-degree monomials and forming $\mathrm{Tr} \circ g : \mathbb{F}_q \to \mathbb{F}_2$ gives the classical *dual BCH codes*, introduced independently by Bose–Ray-Chaudhuri [BRC60] and Hocquenghem [Hoc59]. These codes are linear over $\mathbb{F}_2$, have length q and dimension $\frac{d+1}{2} \log_2 q$, and their distance analysis relies entirely on the Weil bound for additive character sums. Dual BCH codes are remarkable: for small d they are *unmatched* codes in the sense that, without the Weil bounds, there is no known proof that binary codes of their parameters even exist.

Despite this theoretical richness, decoding these codes from a constant fraction of errors was a long-standing open problem. The essential difficulty is that each evaluation $\psi(g(\alpha))$ carries very little information about g — the quadratic character discards the magnitude and retains only whether $g(\alpha)$ is a quadratic residue — so the system of equations one must solve is both underdetermined and nonlinear. Even the zero-error interpolation problem, of recovering g exactly from the values $\psi(g(\alpha))$ for all $\alpha \in \mathbb{F}_q$, had no polynomial-time solution for non-constant d prior to the work presented here.

The goal of this chapter is to present the first polynomial-time algorithms for decoding both code families from a constant fraction of errors, following Kopparty. The algorithms combine two classical ideas: the *Berlekamp–Welch* framework, which uses a high-multiplicity error-locator polynomial to zero out error positions and reduces decoding to a linear system, and the *Stepanov polynomial method*, the elementary approach to the Weil bounds that works by interpolating auxiliary polynomials vanishing with high multiplicity at points of interest. The fusion of these two ideas gives rise to what we call *pseudopolynomials*: high-degree polynomials all of whose Hasse derivatives, when evaluated on $\mathbb{F}_q$, agree with polynomials of much smaller degree. These objects turn out to be the correct algebraic framework for capturing the special structure of character evaluations of polynomials.

§ 1.1 Theory of Pseudopolynomials

1.1.1 Pseudopolynomials

Definition 1.1.1: Pseudoderivative

Let $F(X) \in \mathbb{F}_q[X]$. Define the ℓ-th pseudoderivative *of F, denoted $F_{\langle \ell \rangle}(X)$, to be the unique polynomial of degree at most $q-1$ whose evaluations on $\mathbb{F}_q$ agree with the evaluations of $F^{(\ell)}(X)$. Explicitly:*

$$F_{\langle \ell \rangle}(X) = \left(F^{(\ell)}(X) \bmod \Lambda(X)\right),$$

where $\Lambda(X) = X^q - X$.

Abusing notation, we will sometimes use $F_{\langle \ell \rangle}$ to denote the function $F_{\langle \ell \rangle} : \mathbb{F}_q \to \mathbb{F}_q$, defined by $F_{\langle \ell \rangle}(\alpha) = F^{(\ell)}(\alpha)$.

Definition 1.1.2: Pseudodegree

Let $F(X) \in \mathbb{F}_q[X]$. We define the pseudodegree *of F, denoted $\mathrm{pdeg}(F)$, to be:*

$$\max_{\ell \geq 0} \deg(F_{\langle \ell \rangle}).$$

*We say $F(X)$ is a k-*pseudopolynomial *if $\mathrm{pdeg}(F) \leq k$. When $\deg(F) = d$ we refer to F as a k-*pseudopolynomial of degree *d.*

By definition, we have same basic observations on how pdeg behaves on addition and multiplication of polynomials.

Observation 1.1. *Like normal degree of polynomials* pdeg *behave the similar way, but we have inequality instead of equality. For any $F, G \in \mathbb{F}_q[X]$*

- $\mathrm{pdeg}(F+G) \leq \max(\mathrm{pdeg}(F), \mathrm{pdeg}(G))$.
- $\mathrm{pdeg}(F \cdot G) \leq \mathrm{pdeg}(F) + \mathrm{pdeg}(G)$.

1.1.1.1 Algebraic Characterization

Lemma 1.1.1 Algebraic Characterization of Pseudodegree

Let $F(X) \in \mathbb{F}_q[X]$, and let $F(X) = \sum_{i \geq 0} F_i(X)\,\Lambda(X)^i$, with each $\deg(F_i) < q$, be the base-$\Lambda(X)$ expansion of $F(X)$. Then $\mathrm{pdeg}(F) = \max_{i \geq 0} \deg(F_i)$.

Proof: We expand $F^{(\ell)}$ by applying the product rule for Hasse derivatives to each term of the base-Λ expansion:

$$F^{(\ell)}(X) = \sum_{i \geq 0} \left(F_i \cdot \Lambda^i\right)^{(\ell)}(X) = \sum_{i \geq 0} \left(\sum_{\substack{\ell_0, \ell_1, \ldots, \ell_i \\ \sum \ell_j = \ell}} F_i^{(\ell_0)}(X) \cdot \prod_{j=1}^{i} \Lambda^{(\ell_j)}(X) \right).$$

When we reduce mod $\Lambda(X)$, recall that $\Lambda^{(0)} = \Lambda \equiv 0$, and $\Lambda^{(j)} = 0$ for $j \notin \{0, 1, q\}$. So the only terms surviving mod Λ

are those with $\ell_1, \ldots, \ell_i \in \{1, q\}$, which forces $\ell \geq i$ and $\ell_0 = \ell - (\ell_1 + \cdots + \ell_i) \in [0, \ell - i]$. Collecting these terms:

$$F^{(\ell)}(X) \bmod \Lambda(X) = \sum_{i=0}^{\ell} \sum_{\ell_0=0}^{\ell-i} F_i^{(\ell_0)}(X) \cdot \underbrace{\left(\sum_{\substack{\ell_1,\ldots,\ell_i \in \{1,q\} \\ \ell_1+\cdots+\ell_i = \ell-\ell_0}} \prod_{j=1}^{i} \Lambda^{(\ell_j)}(X) \right)}_{=: C_{i,\ell,\ell_0,q} \in \mathbb{Z}}$$

$$= (-1)^{\ell} F_\ell(X) + \sum_{i=0}^{\ell-1} \sum_{\ell_0=0}^{\ell-i} F_i^{(\ell_0)}(X) \cdot C_{i,\ell,\ell_0,q},$$

where the leading term isolates $i = \ell$, $\ell_0 = 0$, the unique case with $\ell_1 = \cdots = \ell_i = 1$, which contributes the factor $(-1)^{\ell}$.

Forward direction. When $\deg(F_i) \leq k$ for all i, each $F_i^{(\ell_0)}$ has degree at most k, so the entire expression $F^{(\ell)} \bmod \Lambda(X) = F_{\langle \ell \rangle}(X)$ has degree at most k. Since ℓ was arbitrary, $\operatorname{pdeg}(F) \leq k$.

Reverse direction. Rearranging the identity above gives

$$(-1)^{\ell} F_\ell(X) = F_{\langle \ell \rangle}(X) - \sum_{i=0}^{\ell-1} \sum_{\ell_0=0}^{\ell-i} F_i^{(\ell_0)}(X) \cdot C_{i,\ell,\ell_0,q}.$$

We argue by induction on ℓ. The base case $\ell = 0$ gives $F_0 = F_{\langle 0 \rangle}$ directly, so $\deg(F_0) \leq \operatorname{pdeg}(F)$. In the inductive step, assuming $\deg(F_i) \leq \operatorname{pdeg}(F)$ for all $i < \ell$, every term $F_i^{(\ell_0)}$ on the right has degree at most $\operatorname{pdeg}(F)$, so $\deg(F_\ell) \leq \operatorname{pdeg}(F)$ as well. Hence, $\max_{i \geq 0} \deg(F_i) \leq \operatorname{pdeg}(F)$. The two directions together give $\operatorname{pdeg}(F) = \max_{i \geq 0} \deg(F_i)$. ■

1.1.1.2 Multiplicities

Now the pseudopolynomials are special because even though we may have a very large degree pseudopolynomials the multiplicities of an irreducible factor is small. The following lemma gives a bound on the multiplicity modulo q.

Lemma 1.1.2 Multiplicity Bound for Pseudopolynomials

Let $k < q$, and let $F(X)$ be a nonzero k-pseudopolynomial of degree D over $\mathbb{F}_q$. For an irreducible polynomial $H(X) \in \mathbb{F}_q[X]$, let μ be the highest power of $H(X)$ that divides $F(X)$. Then:

$$\mu \bmod q \in \left[0,\ k + \left\lfloor \frac{D}{q} \right\rfloor\right].$$

Proof: Pick a root $\alpha \in \overline{\mathbb{F}_q}$ of $H(X)$, so that $\mu = \operatorname{mult}(F, \alpha)$ and $F^{(\mu)}(\alpha) \neq 0$. By Lemma 1.1.1, the base-Λ expansion of F has the form $F(X) = \sum_{i=0}^{t-1} F_i(X)\Lambda(X)^i$ with $t = \lfloor D/q \rfloor + 1$ and $\deg(F_i) \leq k$.

It suffices to show $F^{(\ell)}(X) = 0$ for every ℓ with $\ell \bmod q \in [k + t, q - 1]$, since this would confine $\mu \bmod q$ to $[0, k + t - 1] = [0, k + \lfloor D/q \rfloor]$.

Expanding via the product rule:

$$F^{(\ell)}(X) = \sum_{i=0}^{t-1} \sum_{\substack{\ell_0,\ell_1,\ldots,\ell_i \\ \sum \ell_j = \ell}} F_i^{(\ell_0)}(X) \cdot \prod_{j=1}^{i} \Lambda^{(\ell_j)}(X).$$

A summand vanishes if $\ell_0 > k$, since $\deg(F_i) \leq k$ forces $F_i^{(\ell_0)} = 0$. It vanishes if some $\ell_j \notin \{0, 1, q\}$, since $\Lambda^{(\ell_j)} = 0$ for such ℓ_j. It also vanishes mod Λ if some $\ell_j = 0$, since that contributes a factor $\Lambda^{(0)} = \Lambda$. Thus, every nonzero summand must have $\ell_0 \in [0, k]$ and $\ell_1, \ldots, \ell_i \in \{1, q\}$.

When $\ell \bmod q \in [k+t, q-1]$, no such decomposition is possible: a sum of one integer from $[0,k]$ and at most $t-1$ integers from $\{1,q\}$ has residue mod q at most $k+(t-1) < k+t$. So every summand vanishes, giving $F^{(\ell)}(X) = 0$. ■

From this lemma we can get a relation of size of a set of zero and the degree of then interpolated pseudopolynomials which becomes zero at the set. For any $S \subseteq \mathbb{F}_q$, and any c, k, M satisfying

$$|S| < \frac{c}{M} \cdot k,$$

there exists a nonzero k-pseudopolynomial $F(X)$ of degree $D < cq$ such that for all $\alpha \in S$, $\mathrm{mult}(F,\alpha) \geq M$. This follows from dimension and constraint counting.

In the reverse direction, the following theorem shows that when q is prime, every nonzero k-pseudopolynomial of degree $D < cq$ has at most $2\frac{c}{M}k$ roots of multiplicity at least M.

Theorem 1.1.3 High Multiplicity Zeroes Bound, [Kop26]

Let q be prime, and let $F(X)$ be a nonzero k-pseudopolynomial of degree $D < cq$ over $\mathbb{F}_q$. Suppose M satisfies $c < M < q$. Then:

$$\left|\{\alpha \in \mathbb{F}_q \mid \mathrm{mult}(F,\alpha) \geq M\}\right| \leq \min\left(\frac{c}{M-c+1} \cdot k + c, \quad k\right).$$

We will not prove this theorem. The proof uses some arguments from [GK14] on wronskian ranks. The full proof of this theorem is in [Kop26, Section 5.7].

1.1.1.3 Codes From Pseudopolynomials

Now using Pseudopolynomials we can create codes similar to univariate multiplicity codes. Let $\Sigma = \mathbb{F}_q^M$. The codewords lie in $\Sigma^{\mathbb{F}_q}$: for each k-pseudopolynomial $F(X)$ of degree $D < cq$, the codeword $\mathrm{Enc}(F) : \mathbb{F}_q \to \Sigma^{\mathbb{F}_q}$ is defined:

$$\mathrm{Enc}(F)(\alpha) = \left(F^{(0)}(\alpha),\ F^{(1)}(\alpha),\ \ldots,\ F^{(M-1)}(\alpha)\right).$$

This code has cardinality $|\Sigma|^{ck/M}$, block length q, rate $R = \frac{c}{M} \cdot \frac{k}{q}$, and minimum distance at least

$$\left(1 - \min\left(\frac{k}{q},\ \frac{c}{M-c+1} \cdot \frac{k}{q} + \frac{c}{q},\ \frac{c}{M}\right)\right) \cdot q,$$

which is always at least $(1-2R) \cdot q$.

1.1.2 Twisted Pseudopolynomials

Definition 1.1.3: r-Twisted Pseudopolynomial

Suppose $r : \mathbb{F}_q \to \mathbb{F}_q$ is a function. Let $k < q$. We say $F(X) \in \mathbb{F}_q[X]$ is an r-twisted (h,M)-pseudopolynomial if for all $\ell < M$, there is some $U_\ell(X) \in \mathbb{F}_q[X]$ of degree at most h such that:

$$F_{\langle \ell \rangle} = r \cdot U_\ell.$$

The following lemma shows that any two r-twisted (h,M)-pseudopolynomials with degree at most cq are very closely related, provided h and c are small enough.

Lemma 1.1.4

Let c, h, M be parameters, with $c < M/2$. Let $r : \mathbb{F}_q \to \mathbb{F}_q$. Suppose $F(X), G(X) \in \mathbb{F}_q[X]$ with $\deg(F), \deg(G) < cq$

are both r-twisted (h, M)-pseudopolynomials and $k > {}^{M}/_{M-2c} \cdot (h+1)$. Then there exist nonzero k-pseudopolynomials $A(X), B(X)$, with $\deg(A), \deg(B) < Mq$, such that:

$$A(X) \cdot F(X) = B(X) \cdot G(X).$$

Proof: Now if $k \geq q$ then the result is immediate. So assume $k < q$. Since F, G both are r-twisted (h, M)-pseudopolynomial for each $\ell < M$, we have:

$$F_{\langle \ell \rangle} = r \cdot U_\ell, \qquad G_{\langle \ell \rangle} = r \cdot V_\ell,$$

where $U_\ell(X), V_\ell(X) \in \mathbb{F}_q[X]$ have degrees at most h.

The goal is to find A and B in the base-Λ form

$$A(X) = \sum_{i=0}^{M-c-1} A_i(X)\, \Lambda(X)^i, \qquad B(X) = \sum_{i=0}^{M-c-1} B_i(X)\, \Lambda(X)^i,$$

with $\deg(A_i), \deg(B_i) \leq k$, so that $AF = BG$. Treating the coefficients of the A_i and B_i as unknowns, their total count exceeds $2(M-c) \cdot k$.

The condition $AF = BG$ is captured by requiring $(A \cdot F)_{\langle \ell \rangle} = (B \cdot G)_{\langle \ell \rangle}$ for each $\ell < M$. Expanding each pseudo-derivative using the twisted conditions on F and G, these equalities can be packaged into a single identity in $\mathbb{F}_q[X, T]$:

$$\left(\sum_{\ell_1 < M} A_{\langle \ell_1 \rangle}(X) \cdot T^{\ell_1} \right) \cdot \left(\sum_{\ell_2 < M} U_{\ell_2}(X) \cdot T^{\ell_2} \right) = \left(\sum_{\ell_3 < M} B_{\langle \ell_3 \rangle}(X) \cdot T^{\ell_3} \right) \cdot \left(\sum_{\ell_4 < M} V_{\ell_4}(X) \cdot T^{\ell_4} \right) \pmod{T^M}.$$

Each of the M coefficient equalities (in $\mathbb{F}_q[X]$, of degree at most $h+k$) is a homogeneous linear condition on the unknowns, giving at most $(h+k+1) \cdot M$ constraints in total. Since by assumption

$$(h+k+1) \cdot M < 2k \cdot (M-c),$$

the number of constraints is strictly less than the number of unknowns, guaranteeing a nonzero solution.

It remains to verify that any such solution satisfies $AF = BG$ as a polynomial identity, not merely on $\mathbb{F}_q$. The construction ensures $(A \cdot F - B \cdot G)_{\langle \ell \rangle} = 0$ for every $\ell < M$, so every point of $\mathbb{F}_q$ is a root of $AF - BG$ of multiplicity at least M. But

$$\deg(A \cdot F - B \cdot G) < cq + (M-c)q = Mq,$$

so the polynomial $AF - BG$ has more roots (counting multiplicity) than its degree, forcing $AF - BG = 0$. ■

The next lemma shows that if we have many r-twisted (h, M)-pseudopolynomials with degree at most cq and c, h small, then they are nontrivially related. The smallness requirement on c is weaker, but the deduced relation is also weaker.

Lemma 1.1.5

Let m, c, h, M be parameters, with $c < {}^{m-1}/_{mM}$. Let $r : \mathbb{F}_q \to \mathbb{F}_q$.

Suppose $F_1(X), \ldots, F_m(X) \in \mathbb{F}_q[X]$ with $\deg(F_i) < cq$ are all r-twisted (h, M)-pseudopolynomials. Let $k > {}^{M}/_{(m-1)M-mc} \cdot (h+1)$. Then there exist k-pseudopolynomials $A_1(X), \ldots, A_m(X)$ with $\deg(A_i) < Mq$ and not all zero, such that:

$$\sum_i A_i(X)\, F_i(X) = 0.$$

The proof is exactly like that of Lemma 1.1.4, which is the $m = 2$ case. The exact same argument also gives robust version.

Theorem 1.1.6 Robust Version for Many

Let m, c, h, γ, M be parameters, with $c < {}^{(m-1)-m\gamma}/_{mM}$. Let $r_1, \ldots, r_m : \mathbb{F}_q \to \mathbb{F}_q$, with:

$$\left|\{\alpha \in \mathbb{F}_q \mid r_i(\alpha) \neq r_j(\alpha) \text{ for some } i, j\}\right| \leq \gamma q.$$

Suppose $F_1(X), \ldots, F_m(X) \in \mathbb{F}_q[X]$ with $\deg(F_i) < cq$ are r_i-twisted (h, M)-pseudopolynomials. Let $k > {}^{M}/_{(m-1-m\gamma)M-mc} \cdot (h+1)$. Then there exist k-pseudopolynomials $A_1(X), \ldots, A_m(X)$, with $\deg(A_i) < Mq$, not all zero, such that:

$$\sum_i A_i(X)\, F_i(X) = 0.$$

§ 1.2 Quadratic Character Evaluations for Odd Characteristic

Let $\psi \in \boldsymbol{\mathcal{X}}_q$ is a non-trivial multiplicative character of $\mathbb{F}_q$ where $\operatorname{ord}(\psi) = d$. Then the polynomial based code we can construct is $(\psi \circ f(\alpha))_{\alpha \in \mathbb{F}_q}$ where $f \in \mathbb{F}_q[X]$. Now suppose g is another polynomial. Then for any $\alpha \in \mathbb{F}_q$,

$$\psi((f \cdot g^d)(\alpha)) = \psi(f(\alpha)) \cdot \psi^d(g(\alpha)) = \psi(f(\alpha))$$

So we assume f doesn't have a power d factor. Similarly, for any α, β where $\beta \neq 0$

$$\psi((\beta^d \cdot f)(\alpha)) = \psi^d(\beta) \cdot \psi(f(\alpha)) = \psi(f(\alpha))$$

Since different leading coefficient can lead to same character evaluation we assume f to be monic.

Now we take the quadratic character $\eta \in \boldsymbol{\mathcal{M}}_q$. Hence, we assume the polynomials are monic and squarefree. Then by Theorem **??** we have

$$\left|\sum_\alpha \eta((g_1 \cdot g_2)(\alpha))\right| < O(d\sqrt{q})$$

Therefore the vectors $\eta \circ g_1$ and $\eta \circ g_2$ differs in at least $\left(q - O(d\sqrt{q})\right) = \left(\frac{1}{2} - \frac{d}{\sqrt{q}}\right) q$ many locations. So their hamming distance is at least $\left(\frac{1}{2} - \frac{d}{\sqrt{q}}\right) q$ which gives the distance of the code.

Algorithm 1: Decoding from $1/8$–th of distance of Quadratic Character Evaluation

Input: Noisy received word, $r : \mathbb{F}_q \to \{0, 1, -1\}$, with degree $d \leq O(\epsilon\sqrt{q})$ and error bound $e \leq (1/8 - \epsilon)\, q$

Output: Find the closest vector $\eta \circ f$ where $f \in \mathbb{F}_q[X]$ of degree at most d

begin

Set $M = 16/\epsilon \cdot d$, $c = M/2$, $h = 2 \cdot e$, $D = d((q-1)/2 + M) + cq$, $u = h + d \cdot M$

Use system of linear equations to find $F(X) \in \mathbb{F}_q^{\leq D}[X]$ and $g_0(X), \ldots, g_{M-1}(X) \in \mathbb{F}_q^{\leq u}[X]$ not all zero such that for all $\alpha \in \mathbb{F}_q$, $i \in \{0, \ldots, M-1\}$:

$$F^{(i)}(\alpha) = r(\alpha) \cdot g_i(\alpha)$$

Complete factorization of $F(X)$ into distinct monic irreducible factors $\{h_1, \ldots, h_k\}$ such that there exists $k_j \in \mathbb{N}$ for all $j \in [k]$ such that

$$F(X) = a \prod_{j=1}^{k} h_j(X)^{k_i}$$

Compute the set $J = \left\{ j \in [t] : k_j \in [3q/8, 7q/8] \right\}$. Then compute $f(X) = \prod_{j \in J} h_j$.

return $f(X)$

Chapter 2

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